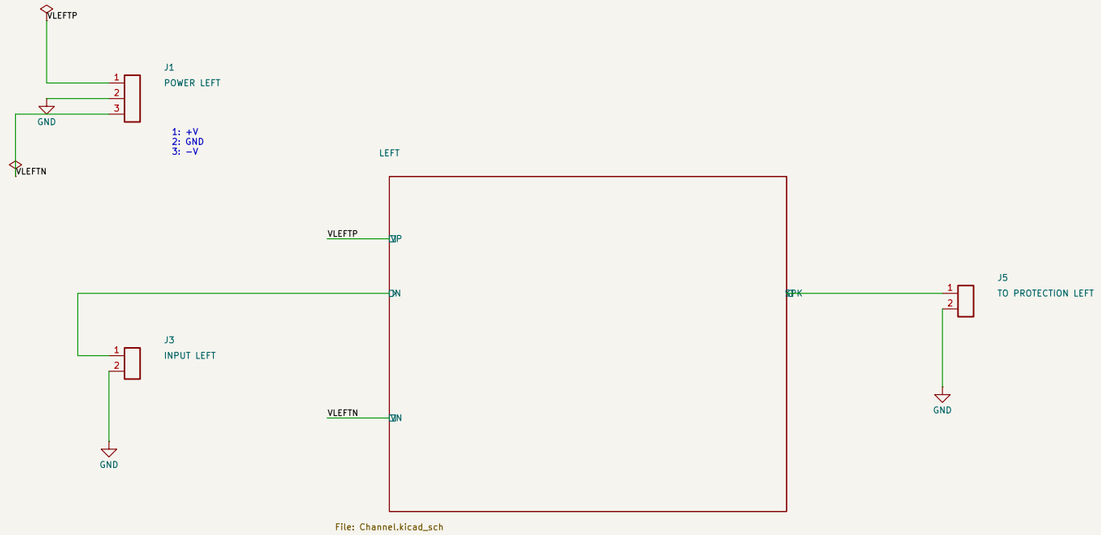
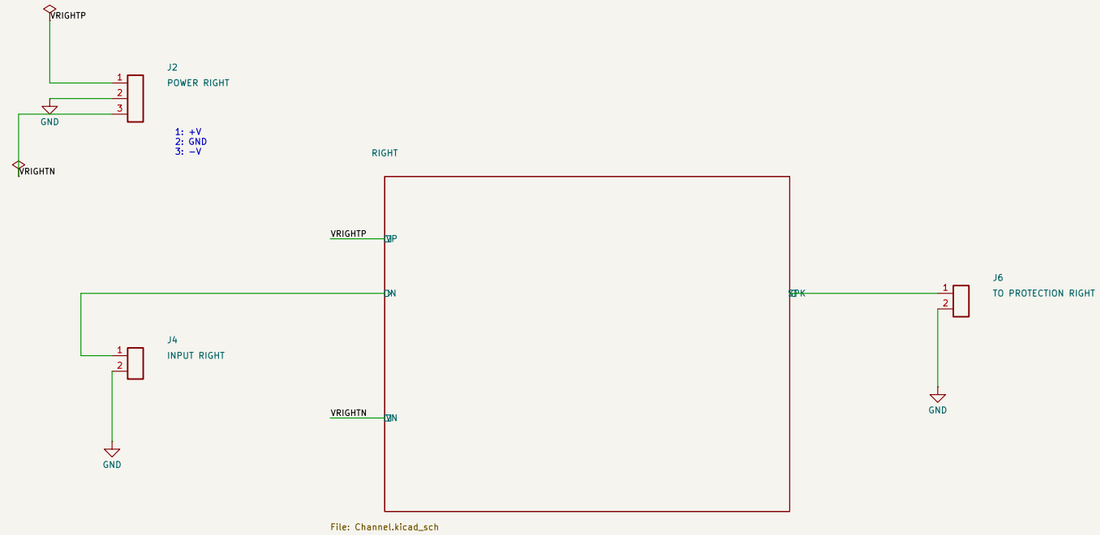
MOSFETAmp2 – STEREO DISCRETE CLASS-AB

REV 0.2: SIMULATION DEVELOPMENT / NOT FABRICATION RELEASED
8 ohm prototype; external current-limited +/-12 to +/-15 V first.
No mains circuitry. External DC speaker protection required. Read README and bench procedure.
BIAS: fixed-ratio sensor + programmed offset; MOSFET VGS slope and SOA still need measurement.





Assembly gates
1. Fit bias short J101/J201; leave R114/R214 unfitted; trimmers at MAXIMUM resistance.
2. Verify live tab insulation and Q108/Q208 thermal contact.
3. Connect supply GND to the junction of the two series bench outputs.
4. Connect oscilloscope earth clips to board GND only.
5. Use dummy loads; no speakers until DC protection and warm-up tests pass.

GND

External PSU assumption envelope: roughly +/-12.5 V valley to +/-20.6 V peak.
Final transformer and capacitors not fixed. Recalculate and measure before use.
Schematic connected and ERC checked separately from electrical/model/thermal validation.
The original empty PCB is retained. Layout is for the user; no routing supplied.

Sheet: /		
File: MOSFETAmp2.kicad_sch		
Title: MOSFETAmp2 stereo Class-AB		
Size: A3	Date: 2026-10-25	**Rev: 0.2 EXPERIMENTAL**
KiCad E.D.A. 10.0.5		Id: 1/3

INPUT / DIFFERENTIAL PAIR

VAS + ACTIVE LOAD

THERMAL BIAS / PROGRAMMED OFFSET

DRIVERS / OUTPUT PAIR

OUTPUT / DECOUPLING

TP104
VP

VP

C105
220u / 35V

C107
100n / 63V

GND

GND

D102
1N4148

D101
1N4148

R107
330R

TREF

Q105
BC556

R108
12k

VN

R111
100R

TREF

Q107
BC556

BP

Q109
BD139

D109
1N4148

D112
1N4148

C103
100n / 63V

Q108
BD139

R112
3k9

VP

R127
680R

Q117
BC556

TREF

R117
100R

Q111
IRF540N

R121
47k

D110
BZX85C10

R115
2k2

TP101
SN

R119
OR33 / 5W

J101
BIAS SHORT – FIT FIRST

BP

BN

R106
100R

R105
100R

Q101
BC556

R113
1k SLOPE SEL

Q116
BC546

R126
10R / 2W
L101

0.7uH AIR / 3A

TP102
OUT

OUT

SPK

R101
1k

GIN

B1

Q102
BC556

FB

R103
22k

OUT

R102
22k

C101
220p C0G

GND

GND

R104
2k2

GND

C102
330p C0G

R114
1k SOT

D113
LED RED

RV101
200R MULTITURN

VN

D107
1N4148

D108
BAT46

C2N

VP

Q115
BC546

BN

R125
10R / 2W

R120
OR33 / 5W

C104
100n FILM / 63V

TP103
SP

GND

R116
2k2

R122
47k

D111
BZX85C10

R118
100R

Q112
IRF9540N

Input overload clamps

D105
1N4148

FB

B1

D106
1N4148

Q104
BC546

Q103
BC546

Q106
BD139

R110
10k

R109
100R

Q110
BD140

GND

GND

C106
220u / 35V

C108
100n / 63V

VN

VN

TP105
VN

Q101/Q102: match VBE at 1 mA; bond flat faces.
R101 + R102: input about 23k; gain 11 from B1.
about 10.52 (20.44 dB) at IN. DC coupled.
C102 = 330p C0G 50V radial; D107 + D108
prevent VAS saturation. VAS/load about 6 mA.

BIAS: Q108 on heatsink beside its MOSFETs.
k = 1 + 3k9/R113 (R113 = slope select).
D109/D112 on PCB beside Q109/Q110.
Offset = 3k9 x (VD113) – VBE)/(R114 + RV101).
R114 unfitted / RV101 max resistance = MIN BIAS.
Fit J101 for first power–up. See BIAS_STUDY.md.

16.5 mV across each OR33 = 50 mA (TP101–TP102, TP102–TP103).
OUT is feedback sense BEFORE the coil.
MOSFET drain tabs are live: INSULATE.
Gate Zeners: N K to gate; P K to source.
8 ohm prototype: no short–circuit protection;
external DC speaker protection required.

Sheet: /LEFT/
File: Channel.kicad_sch

Title: Class–AB channel – experimental Rev 0.2

Size: A3 | Date: 2026–10–25 | **Rev: 0.2 EXPERIMENTAL**

KiCad E.D.A. 10.0.5 | Id: 2/3

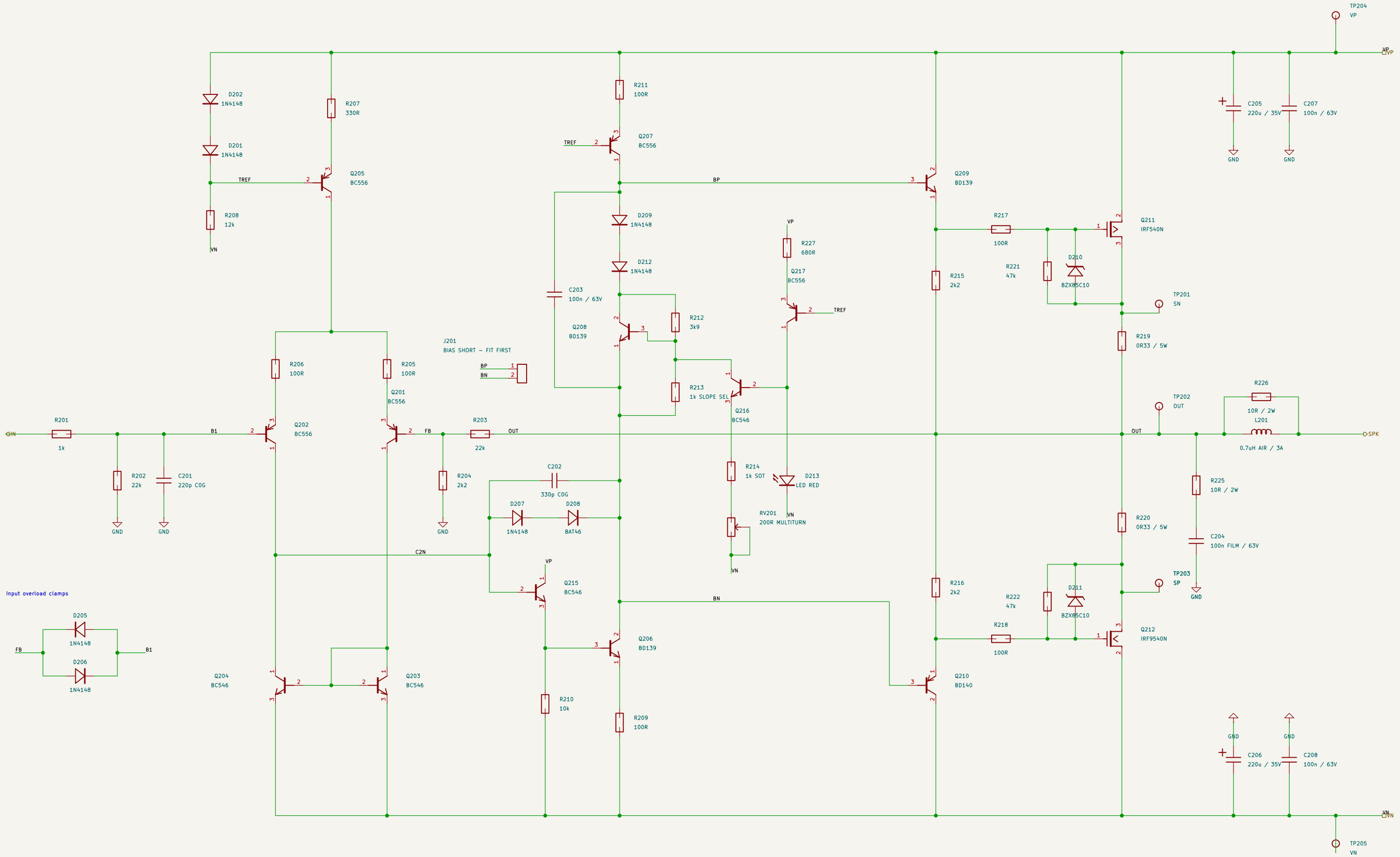
INPUT / DIFFERENTIAL PAIR

VAS + ACTIVE LOAD

THERMAL BIAS / PROGRAMMED OFFSET

DRIVERS / OUTPUT PAIR

OUTPUT / DECOUPLING



Q101/Q102: match VBE at 1 mA; bond flat faces.
R101 + R102: input about 23k; gain 11 from B1.
about 10.52 (20.44 dB) at IN. DC coupled.
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D109/D112 on PCB beside Q109/Q110.
Offset = 3k9 x (VD113) – VBE)/(R114 + RV101).
R114 unfitted / RV101 max resistance = MIN BIAS.
Fit J101 for first power–up. See BIAS_STUDY.md.

16.5 mV across each 0R33 = 50 mA (TP101–TP102, TP102–TP103).
OUT is feedback sense BEFORE the coil.
MOSFET drain tabs are live: INSULATE.
Gate Zeners: N K to gate; P K to source.
8 ohm prototype: no short–circuit protection;
external DC speaker protection required.

Sheet: /RIGHT/
File: Channel.kicad_sch

Title: Class–AB channel – experimental Rev 0.2

Size: A3	Date: 2026–10–25	**Rev: 0.2 EXPERIMENTAL**
KiCad E.D.A. 10.0.5		Id: 3/3